Tuesday, 29 September 2026 8:01 PM

(1) 2 fair dice that are rolled (i,j). $\Omega = \{(i,j) : 1 \le i \le 6, 1 \le j \le 6\}$

$A = \{(i,j) : i+j = 7\}$

$B = \{(i,j) : i \le 3\}$

(a) Find $P(A \cup B)$

(b) Are A, B independent?

Fair dice (i,j) are all equally likely

$|\Omega| = 36$, $P(E) = \frac{|E|}{36}$

(a) $P(A \cup B) = \frac{|A \cup B|}{36} = \frac{21}{36} = \frac{7}{12}$

$A = \{(i,j) : i+j=7\} = \{\{1,6\},\{2,5\},\{3,4\},\{4,3\},\{5,2\},\{6,1\}\}$

$|A| = 6$

$B = \{(i,j) : i \le 3\} = \{\{1,1\},\{1,2\},\ldots,\{1,6\},\{2,1\},\{2,2\},\ldots,\{2,6\},\{3,1\},\{3,2\},\ldots,\{3,6\}\}$

$|B| = 18$

$|A \cup B| = |A| + |B| - |A \cap B| = 6 + 18 - 3 = 21$

$A \cap B = \{\{1,6\},\{2,5\},\{3,4\}\}$ $|A \cap B| = 3$

(b) A, B are ind? ✓

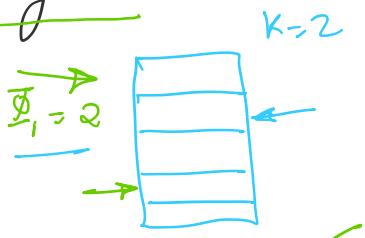
$P(A \cap B) \overset{\checkmark}{=} P(A) \cdot P(B)$

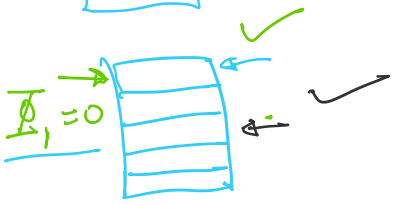
$\frac{3}{36} \quad \frac{6}{36} \quad \frac{18}{36}$

$\frac{1}{12} = \frac{1}{12}$

(2) A dataset D containing n distinct points.

Data scientist selects K data points ind. with replacement. Let Φ_1 denote the number of distinct data pts. that appear exactly once in the selection.





(a) Suppose $K = 2$

(1) Find PMF of Φ_1 ✓

(2) Find $\mathbb{E}[\Phi_1]$ ✓

- $\Phi_1 = 2$ $\quad$ $\Phi_1 = 0$

$$P(\Phi_1 = 0) = \frac{1}{n}, \quad P(\Phi_1 = 2) = \frac{n-1}{n}$$

- $$\mathbb{E}[\Phi_1] = 2 \times \frac{(n-1)}{n} + 0 \times \frac{1}{n} = \frac{2(n-1)}{n}$$

$\frac{n \cdot (n-1)}{n^2}$

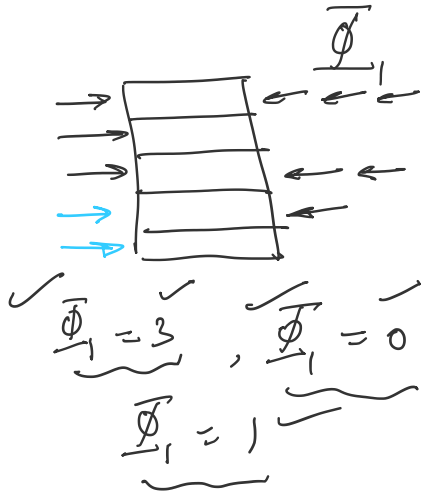
$K = 3$.

(i, j)

$\rightarrow n$

$\rightarrow n^2$ $\quad = \frac{1}{n}$

$(1,1)$
$(2,2)$
$\vdots$
(n,n)



n^3

Eg E-commerce company

$X \triangleq$ # promotional emails opened

Y = # purchases made

Joint pmf $p(x=x, y=y) \leftarrow P_{XY}$

	$Y=0$	$Y=1$	$Y=2$
$X=0$	.2	.05	.05
$X=1$	.1	.2	.1
$X=2$	.05	.1	.15

(a) P_X

$P(X=0) = .3$

$P(X=1) = .4$

$P(X=2) = .3$

(b)

$P(Y=0) = .35$

$P(Y=1) = .35$

$P(Y=2) = .3$

(c) $P(Y=2 \mid X=2)$

$= \dfrac{P(Y=2, X=2)}{P(X=2)} = \dfrac{.15}{.3} = \dfrac{1}{2}$

(d) Joint CDF $F_{XY}(\alpha, \beta) = P(X \le \alpha, Y \le \beta)$

$F_{XY}(1,2) = P(X \le 1, Y \le 2)$

$X \backslash Y$	0	1	2
0	.2	.05	.05
1	.1	.2	.1
2	.05	.1	.15

$\text{Var}(Y) = 1.55 - (.95)^2 \approx .65$

$P_X = \{ \underset{0}{.3} \quad \underset{1}{.4} \quad \underset{2}{.3} \}$

$P_Y = \{ \underset{0}{.35} \quad \underset{1}{.35} \quad \underset{2}{.3} \}$

$\rho_{XY} = \dfrac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(X)}}$

$= \dfrac{\mathbb{E}[XY] - \mu_X \cdot \mu_Y}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}$

$\mu_X = 0 \times .3 + 1 \times .4 + 2 \times .3 = 1$

$\mathbb{E}(X^2) = 0 \times .3 + 1 \times .4 + 4 \times .3 = 1.6$

$\mu_Y = 0 \times .35 + 1 \times .35 + 2 \times .3 = .95$

$\mathbb{E}(Y^2) = 0 \times .35 + 1 \times .35 + 4 \times .3 = 1.55$

$\text{Var}(X) = \mathbb{E}(X^2) - \mu_X^2 = 1.6 - 1 = .6$

$$E(XY) = \sum_x \sum_y x \cdot y \cdot P_{XY}(x,y)$$

$$= 1 \cdot 1 \cdot P(1,1) + 1 \cdot 2 \cdot P(1,2)$$
$$+ 2 \cdot 1 \, P(2,1) + 2 \cdot 2 \, P(2,2)$$
$$= .2 + .2 + .2 + .6$$
$$= 1.2$$

$$\rho_{XY} = \frac{E(XY) - \mu_X \mu_Y}{\sqrt{\text{var}(X)}\sqrt{\text{var}(Y)}} = \frac{1.2 - 1 \times .95}{\sqrt{.6} \times \sqrt{.65}} \approx .4$$

$$\text{Var}(X) = E\left((X - \mu_X)^2\right) \quad (X-\mu_X)^2$$

$(X-\mu_X)^2 \rightarrow$ $X=0 \rightarrow 1 \rightarrow .3$

$X=1 \rightarrow 0 \rightarrow .4$

$X=2 \rightarrow 1 \rightarrow .3$

X, Y ind or not ?

↗ stricter

↘ $P_{XY}(x,y) \stackrel{?}{=} P_X(x) \cdot P_Y(y)$

↘ X, Y uncorrelated or not ?

$E(X \cdot Y) = \mu_X \cdot \mu_Y$ $\quad E(XY) = 0$ ↓ orthogonal

or $\text{cov}(X,Y) = 0$?

$P_{XY}(1,2) = P_X(1) \cdot P_Y(2)$

$P_{XY}(1,1) \stackrel{?}{=} P_X(1) \cdot P_Y(1)$?

Team A, Team B